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实验室名称：国家电器产品质量监督检验中心
Lab Name: China National Center for Quality Supervision and Test of Electrical Apparatus Products

№ 21M2078-S

检验(试验)报告
Test Report

委托单位：Fuzhou Tianyu Electric Co., Ltd.
Client:

产品名称：Power transformer
Name of Product:

产品型号：SZ22-50000/110-NX1
Product Type:

检验类别：Commission test
Test Category:

DQJC

China National Center for Quality Supervision and Test of Electrical Apparatus Products

Test Report

№: 21M2078-S

Client	Fuzhou Tianyu Electric Co., Ltd.	Test category	Commission test
Manufacturer	Fuzhou Tianyu Electric Co., Ltd.	Date of sample receiving	/
Name of sample	Power transformer	Type of sample	SZ22-50000/110-NX1
Address of manufacturer	No. 28, Yaoxi Road, Nanyu Town, Minhou County, Fuzhou City, Fujian Province	Original number or date of production	AN17801
Date of test	From Aug. 10, 2021 to Aug. 17, 2021	Number of sample	1 set
Test items	Routine test Type test (including calculation of the winding hot-spot temperature-rise) Determination of flash point (closed cup) in insulating liquid Measurement of bushing capacitances and dielectric dissipation factor (tanδ) Measurement of frequency response Measurement of no-load excitation characteristics Long-duration no-load test Measurement of zero-sequence impedances on three-phase transformers Measurement of the harmonics of the no-load current	Test standards	GB/T 1094.1—2013 GB/T 1094.2—2013 GB/T 1094.3—2017 GB/T 1094.10—2003 GB/T 6451—2015 GB/T 7595—2017 JB/T 10088—2016 GB 20052—2020 IEC 60076-1: 2011 IEC 60076-2: 2011 IEC 60076-3: 2013+AMD1: 2018 IEC 60076-10: 2016 Commission requirements
Test conclusion	The test results of routine test, type test (including calculation of the winding hot-spot temperature-rise), determination of flash point (closed cup) in insulating liquid, measurement of bushing capacitances and dielectric dissipation factor (tanδ), measurement of frequency response, measurement of no-load excitation characteristics, long-duration no-load test, measurement of zero-sequence impedances on three-phase transformers and measurement of the harmonics of the no-load current of power transformer (type: SZ22-50000/110-NX1) are in accordance with test standards and commission requirements. The sample has passed the above tests. Signing and issuing date: 2021.09.14 Note: the conclusion is valid only for the inspected and tested sample.		
Remarks	All tests were performed in Fuzhou Tianyu Electric Co., Ltd.		

Compiled by: 丁大为 Proofread by: 廖如[illegible] Checked by: 王占宾 Approved by: [illegible]

1. Sample parameters

Rated power: 50000kVA

Rated voltage: 110/10.5kV

Rated current: 262.4/2749.4A

Rated frequency: 50Hz

Number of phases: 3

Tapping ranges: ±8×1.25%

Connection symbol: YNd11

Cooling method: ONAN

Class of insulation and heat-resistant: A

Insulation level:	HV	Um/LI/LIC/AC	126/480/530/200kV
	HVN	Um/LI/AC	72.5/325/140kV
	LV	Um/LI/LIC/AC	12/75/85/35kV

2. Test standards

GB/T 1094.1—2013 *Power transformers—Part 1: General*

GB/T 1094.2—2013 *Power transformers—Part 2: Temperature rise for liquid-immersed transformers*

GB/T 1094.3—2017 *Power transformers—Part 3: Insulation levels, dielectric tests and external clearances in air*

GB/T 1094.10—2003 *Power transformers—Part 10: Determination of sound levels*

GB/T 6451—2015 *Technical parameters and requirements of oil-immersed power transformer*

GB/T 7595—2017 *Quality of transformer oils in service*

JB/T 10088—2016 *Sound level for 6kV-1000kV power transformers*

GB 20052—2020 *Minimum allowable values of energy efficiency and the energy efficiency grades for power transformers*

IEC 60076-1: 2011 *Power transformers—Part 1: General*

IEC 60076-2: 2011 *Power transformers—Part 2: Temperature rise for liquid-immersed transformers*

IEC 60076-3: 2013+AMD1: 2018 *Power transformers—Part 3: Insulation levels, dielectric tests and external clearances in air*

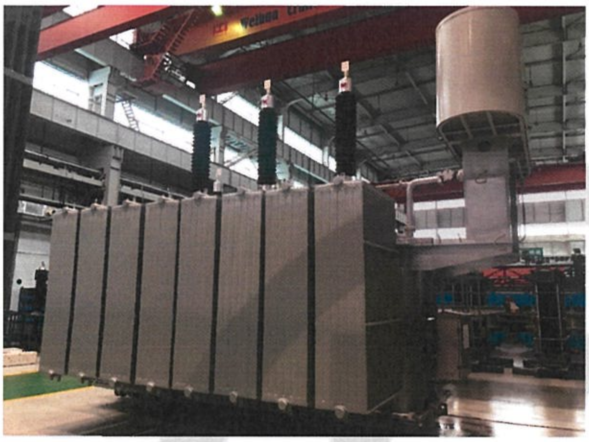
IEC 60076-10: 2016 *Power transformers—Part 10: Determination of sound levels*

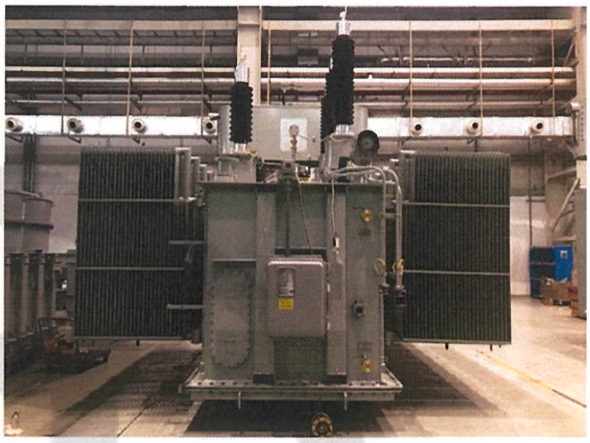
Commission requirements

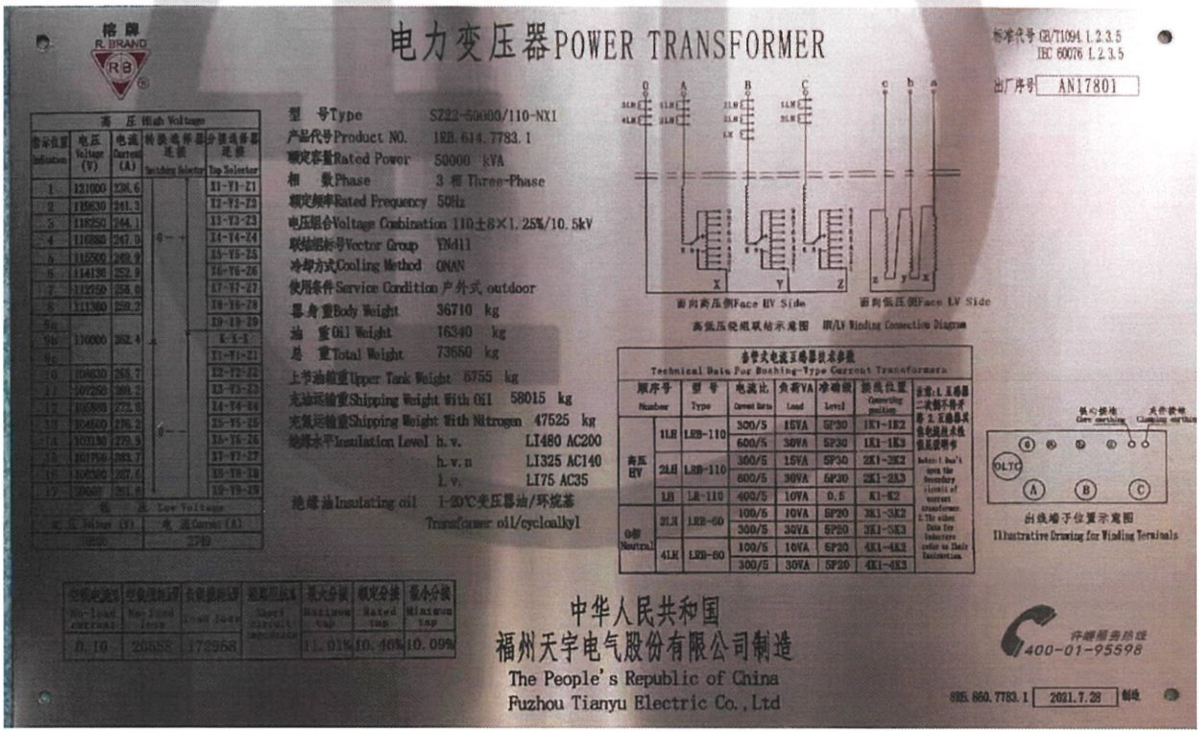
3. Sample description

The power transformer is for outdoor use. Loss parameters meet the requirements of energy efficiency grade 1 of GB 20052—2020, external photos of the sample have been attached.

Photos of the sample







Summary of test results

No	Test items	Specified value Standards (commission requirements)	Measured value	Results
1	Measurement of d.c. insulation resistance between each winding to earth and between windings (routine test)	Providing value of insulation resistance (GΩ): ＞6.0 Providing absorption ratio: ＞1.3	See 4.1	PASS
2	Check of core and frame insulation for liquid-immersed transformers with core or frame insulation (routine test)	Providing value of insulation resistance at 20°C (GΩ): ≥1.5	See 4.2	PASS
3	Measurement of dissipation factor (tanδ) of the insulation system capacitances (routine test)	Providing dielectric dissipation factor tanδ at 20°C: tanδ: ＜0.5%	See 4.3	PASS
4	Determination of capacitances windings-to-earth and between windings (routine test)	Providing value of capacitance	See 4.4	/
5	Measurement of bushing capacitances and dielectric dissipation factor (tanδ) (commission test)	Providing value of capacitance Providing dielectric dissipation factor tanδ: ＜0.4%	See 4.5	PASS
6	Check of the ratio and polarity of built-in current transformers (routine test)	Providing the value of ratio and polarity relation	See 4.6	PASS
7	Insulation test of auxiliary wiring (AuxW) (routine test)	Wiring for auxiliary power and control circuit: 2.0kV 60s Wiring for secondary winding of current transformer: 2.5kV 60s	2.0kV 60s 2.5 kV 60s	PASS
8	Measurement of voltage ratio and check of phase displacement (routine test)	Voltage ratio tolerance of principal tapping: obtaining the lower of the following values between ±0.5% of declared ratio and ±1/10 of the actual percentage impedance Connection symbol: YNd11	See 4.8	PASS
9	Measurement of winding resistance (routine test)	Maximum resistance unbalance rate Phase resistance: ≤2% Line resistance: ≤1%	HV(phase): 1.07% LV(line): 0.52%	PASS

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<table>
<tr><th rowspan="2">No</th><th rowspan="2">Test items</th><th colspan="3">Specified value</th><th colspan="3" rowspan="2">Measured value</th><th rowspan="2">Results</th></tr>
<tr><th colspan="3">Standards
(commission requirements)</th></tr>
<tr><td>10</td><td>Measurement of no-load loss and current
(routine test)</td><td colspan="3">I_0(%): ≤0.30
P_0(kW): 21.000 +0%</td><td colspan="3">0.09
20.468</td><td>PASS</td></tr>
<tr><td>11</td><td>Measurement of no-load loss and current at 90% and 110% of rated voltage
(routine test)</td><td colspan="3">I_0(%): measured
P_0(kW): measured</td><td colspan="3">90% 110%
0.05 0.51
14.719 34.244</td><td>/</td></tr>
<tr><td>12</td><td>Measurement of short-circuit impedance and load loss
(routine test)</td><td colspan="3">t: 75°C
Z(%): 10.5 ±3%
P_k(kW): 175.000 +0%
P_{total} (kW): 196.000 +0%</td><td colspan="3">10.46
172.957
193.426</td><td>PASS</td></tr>
<tr><td>13</td><td>Tests on on-load tap-changers
(routine test)</td><td colspan="3">According to clause 11.7 of GB/T 1094.1-2013, according to clause 11.7 of IEC 60076-1: 2011</td><td colspan="3">Meet the requirements</td><td>PASS</td></tr>
<tr><td rowspan="2">14</td><td rowspan="2">Lightning impulse test
(LI, LIC, LIN)
(routine test, type test)</td><td colspan="3">HV:
Full wave (kV): 480 ±3%
Chopped wave (kV): 530 ±3%
Neutral point (kV): 325 ±3%</td><td colspan="3">476.39～481.37
525.29～529.74
325.57～327.16</td><td rowspan="2">PASS</td></tr>
<tr><td colspan="3">LV:
Full wave (kV): 75 ±3%
Chopped wave (kV): 85 ±3%</td><td colspan="3">74.14～76.43
82.77～85.29</td></tr>
<tr><td>15</td><td>Applied voltage test (AV)
(routine test)</td><td colspan="3">HV neutral point: 140kV 60s
LV: 35kV 60s</td><td colspan="3">140.0kV 60s
35.0kV 60s</td><td>PASS</td></tr>
<tr><td rowspan="4">16</td><td rowspan="4">Line terminal AC withstand test (LTAC)
(routine test)</td><td colspan="6">Test on phase to earth</td><td rowspan="4">PASS</td></tr>
<tr><td colspan="2">Applied voltage (kV):</td><td colspan="2">ac: 21.0</td><td>ab: 21.0</td><td>bc: 21.0</td></tr>
<tr><td colspan="2">Induced voltage (kV): 200</td><td colspan="2">A: 200.0</td><td>B: 200.0</td><td>C: 200.0</td></tr>
<tr><td colspan="2">Duration (s): 120(f_n/f)
Frequency (Hz): ＞50</td><td colspan="4">30
200</td></tr>
</table>

No	Test items	Specified value Standards (commission requirements)	Measured value	Results
17	Induced voltage withstand test and induced voltage test with partial discharge measurement (IVW, IVPD) (routine test)	Three-phase applied voltage		PASS
		$0.4Ur/\sqrt{3}$ (kV) Discharge magnitude ≤50pC	25.4 A: ＜30; B: ＜30; C: ＜30	
		$1.2Ur/\sqrt{3}$ (kV) Duration (min): 1 Discharge magnitude ≤100pC	76.2 1 A: ＜40; B: ＜40; C: ＜40	
		$1.58Ur/\sqrt{3}$ (kV) Duration (min): 5 Discharge magnitude ≤100pC	100.3 5 A: ＜50; B: ＜70; C: ＜50	
		$2.0Ur/\sqrt{3}$ (kV) Duration (min): 0.5	127.0 0.5	
		$1.58Ur/\sqrt{3}$ (kV) Duration (min): 60 Discharge magnitude ≤90pC and the increment ≤50pC	100.3 60 A: ＜50; B: ＜70; C: ＜50	
		$1.2Ur/\sqrt{3}$ (kV) Duration (min): 5 Discharge magnitude ≤100pC	76.2 5 A: ＜30; B: ＜30; C: ＜30	
		$0.4Ur/\sqrt{3}$ (kV) Duration (min): 1 Discharge magnitude ≤50pC	25.4 1 A: ＜20; B: ＜20; C: ＜20	
		Frequency (Hz): ＞50	200	
18	Insulating liquid test, measurement of dissolved gasses in dielectric liquid from each separate oil compartment except diverter switch compartment (routine test, type test, commission test)	Breakdown voltage (kV): ≥60 tanδ(90°C): ≤0.3% Water content (mg/L): ≤10	64.3 0.09% 6.1	PASS
		Providing gas chromatograph analysis: Hydrogen: ＜10μL/L Acetylene: 0 Total hydrocarbon: ≤10μL/L	See 4.18	
		Flash point (closed cup) (°C): ＞170	173.0	
19	Measurement of frequency response (special test)	Providing frequency response characteristics curve	See 4.19	PASS

No	Test items	Specified value Standards (commission requirements)	Measured value	Results
20	Measurement of no-load excitation characteristics (commission test)	Providing no-load excitation characteristic curve	See 4.20	/
21	Long-duration no-load test (commission test)	Applied voltage (kV): 1.1Ur Duration (h): 12 Without acetylene in oil	1.1Ur 12 For gas chromatograph analysis, see 4.18	PASS
22	Measurement of zero-sequence impedances on three-phase transformers (special test)	Providing zero-sequence impedances values (Ω)	See 4.22	/
23	Measurement of the harmonics of the no-load current (commission test)	Providing harmonics of the no-load current of each phase	Harmonics of the no-load current of I_1-I_{19}	/
24	Temperature-rise test (including calculation of the winding hot-spot temperature-rise) (type test)	Top oil temperature-rise limits (K): 53 Winding temperature-rise limits (K): 60 Winding hot-spot temperature-rise limits (K): 73 Temperature-rise of tank and structural parts (K): 70	Top oil temperature-rise: 47.3 HV winding temperature-rise: 54.2 LV winding temperature-rise: 55.3 HV winding hot-spot temperature-rise: 69.7 LV winding hot-spot temperature-rise: 70.5 Temperature-rise limits of tank and structural parts: 50.1	PASS
25	Leak testing with pressure for liquid-immersed transformers (routine test)	Body: Applied pressure (kPa): 30 Duration (h): 24 No oil leakage or damage	30.0 24 No oil leakage or damage	PASS
		On-load tap-changer tank: Applied pressure (kPa): 30 Duration (h): 24 No oil leakage or damage	30.0 24 No oil leakage or damage	
26	Determination of sound levels (type test)	Sound pressure level $\overline{L_{PA}}$ dB(A): ≤58 Sound power level L_{WA}dB(A): ≤80	56 76	PASS
27	Mechanical strength test of tank (type test)	Applied vacuum degree (kPa): 0.133 Applied positive pressure (kPa): 100 Elastic deformation of tank wall (mm): ≤24 Elastic deformation of tank cover (mm): ≤30 Permanent deformation of tank wall (mm): ≤12 Permanent deformation of tank cover (mm): ≤10 No damage	See 4.27	PASS

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4. Test items and results

4.1 Measurement of d.c. insulation resistance between each winding to earth and between windings (routine test)

Test date: Aug. 13, 2021
Relative humidity: 60%; Oil temperature: 32.0°C

Measured parts	R_{15} (GΩ)	R_{60} (GΩ)	Absorption ratio (R_{60}/R_{15})
HV—LV and earth	10.40	13.70	1.31
LV—HV and earth	7.82	11.20	1.43
HV, LV—earth	8.30	11.00	1.32
HV—LV	8.40	13.40	1.59

4.2 Check of core and frame insulation for liquid-immersed transformers with core or frame insulation (routine test)

Test date: Aug. 13, 2021
Relative humidity: 60%; Oil temperature: 32.0°C

Measured parts	Measured insulation resistance (GΩ)	Corrected insulation resistance to 20°C (GΩ)
Core—earth	2.14	3.48
Frame—earth	1.73	2.81

4.3 Measurement of dissipation factor (tanδ) of the insulation system capacitances (routine test)

Test date: Aug. 13, 2021
Relative humidity: 60%; Oil temperature: 32.0°C

Measured parts	Dielectric dissipation factor tanδ (%)	Dielectric dissipation factor tanδ corrected to 20°C (%)	Capacitance (pF)
HV—LV and earth	0.30	0.22	9227
LV—HV and earth	0.33	0.24	12820
HV, LV—earth	0.40	0.29	11710
HV—LV	0.30	0.22	4830

4.4 Determination of capacitances windings-to-earth and between windings (routine test)

Test date: Aug. 13, 2021

See 4.3.

4.5 Measurement of bushing capacitances and dielectric dissipation factor (tanδ) (commission test)

Test date: Aug. 13, 2021

Relative humidity: 60%; Oil temperature: 32.0°C

Measured contents	Applied voltage	Test parts			
		A	B	C	O
tanδ(%)	10kV	0.26	0.33	0.23	0.23
Measured capacitance (pF)		322.3	319.2	313.3	371.4
Nominal capacitance (pF)		325	325	319	375

4.6 Check of the ratio and polarity of built-in current transformers (routine test)

Test date: Aug. 13, 2021

Installation position	Type	Terminal	Standard ratio	Measured ratio	Polarity
A	LRB-110	1K1-1K2	300/5	59.87	-
		1K1-1K3	600/5	120.57	-
		2K1-2K2	300/5	59.23	-
		2K1-2K3	600/5	120.13	-
B	LRB-110	1K1-1K2	300/5	59.91	-
		1K1-1K3	600/5	120.17	-
		2K1-2K2	300/5	59.87	-
		2K1-2K3	600/5	120.21	-
	LR-110	K1-K2	400/5	80.10	-
C	LRB-110	1K1-1K2	300/5	60.12	-
		1K1-1K3	600/5	120.25	-
		2K1-2K2	300/5	59.73	-
		2K1-2K3	600/5	119.47	-
O	LRB-66	3K1-3K2	100/5	20.20	-
		3K1-3K3	300/5	60.27	-
		4K1-4K2	100/5	20.13	-
		4K1-4K3	300/5	60.21	-

4.7 Insulation test of auxiliary wiring (AuxW) (routine test)

Test date: Aug. 13, 2021

Relative humidity: 60%; Ambient temperature: 31.7°C; Air pressure: 102kPa

Parts of applied voltage		Test voltage (kV)	Test duration (s)	Result
Wiring for auxiliary power and control circuit	Powering for on-load tap-changer	2.0	60	PASS
Wiring for secondary winding of current transformer	A	2.5	60	
	B	2.5	60	
	C	2.5	60	
	O	2.5	60	

4.8 Measurement of voltage ratio and check of phase displacement (routine test)

Test date: Aug. 13, 2021

HV winding		LV winding		Transformer ratio by calculation	Measured voltage ratio tolerance (%)			Connection symbol
Tapping position	Voltage (kV)	Tapping position	Voltage (kV)		AB/ab	BC/bc	CA/ca	
1	121.000	/	10.5	11.524	-0.09	-0.09	-0.10	YNd11
2	119.630			11.394	+0.06	+0.06	+0.06	
3	118.250			11.262	+0.06	+0.06	+0.06	
4	116.880			11.132	+0.05	+0.05	+0.05	
5	115.500			11.000	+0.04	+0.04	+0.04	
6	114.130			10.870	+0.02	+0.02	+0.03	
7	112.750			10.738	+0.02	+0.05	+0.01	
8	111.380			10.608	0.00	0.00	+0.01	
9b	110.000			10.476	0.00	0.00	0.00	
10	108.630			10.346	+0.16	+0.16	+0.16	
11	107.250			10.214	+0.16	+0.16	+0.16	
12	105.880			10.084	+0.16	+0.15	+0.15	
13	104.500			9.952	+0.15	+0.15	+0.15	
14	103.130			9.822	+0.14	+0.14	+0.14	
15	101.750			9.690	+0.14	+0.13	+0.14	
16	100.380			9.560	+0.13	+0.12	+0.12	
17	99.000			9.429	+0.12	+0.12	+0.12	

4.9 Measurement of winding resistance (routine test)

Test date: Aug. 13, 2021
Oil temperature: 32.0°C

Winding	Tapping position	Measured values (Ω)			Resistance unbalance rate (%)
		A～O a～b	B～O b～c	C～O c～a	
HV	1	0.3517	0.3534	0.3551	0.96
	2	0.3465	0.3479	0.3497	0.92
	3	0.3398	0.3412	0.3425	0.79
	4	0.3333	0.3346	0.3362	0.87
	5	0.3266	0.3282	0.3296	0.91
	6	0.3201	0.3218	0.3233	1.00
	7	0.3155	0.3152	0.3168	0.56
	8	0.3072	0.3087	0.3105	1.07
	9b	0.2999	0.3004	0.3008	0.30
	10	0.3063	0.3074	0.3090	0.89
	11	0.3126	0.3139	0.3152	0.83
	12	0.3192	0.3203	0.3219	0.84
	13	0.3257	0.3270	0.3280	0.70
	14	0.3324	0.3337	0.3347	0.69
	15	0.3388	0.3402	0.3414	0.82
	16	0.3456	0.3469	0.3479	0.66
	17	0.3524	0.3534	0.3545	0.59
LV	/	4.791×10^{-3}	4.797×10^{-3}	4.816×10^{-3}	0.52

4.10 Measurement of no-load loss and current (routine test)

Test date: Aug. 13, 2021

Voltage multiple	Applied voltage (kV)		No-load current		No-load loss (kW)	
	Average value	r.m.s. value	(A)	(%)	Measured value	Corrected value
90%Ur	9.451	9.447	1.43	0.05	14.713	14.719
100%Ur	10.500	10.509	2.35	0.09	20.486	20.468
110%Ur	11.550	11.706	13.98	0.51	34.713	34.244

Remarks: at 100%Ur, the difference between r.m.s voltage and average voltage is within 3%.

4.11 Measurement of no-load loss and current at 90% and 110% of rated voltage (routine test)
See 4.10.

4.12 Measurement of short-circuit impedance and load loss (routine test)

Test date: Aug. 13, 2021

Oil temperature: 32.0°C

Winding	Tapping position	Applied current I		Measured voltage (kV)	Short-circuit impedance (for each phase)		Load loss (kW)	Total loss (kW)
		(A)	I/Ir (%)		HV impedance (Ω)	(%)	Corrected value	Corrected value
					t=75°C I=Ir	t=75°C I=Ir	t=75°C I=Ir	t=75°C I=Ir
HV-LV	1	128.64	53.92	7.185	32.24	11.01	174.153	194.622
	9b	142.89	54.45	6.265	25.31	10.46	172.957	193.426
	17	162.82	55.85	5.576	19.78	10.09	190.892	211.360

4.13 Tests on on-load tap-changers (routine test)

Test date: Aug. 13, 2021

Operation tests on on-load tap-changers:

a. with the transformer de-energized, eight complete cycles of operation (a cycle of operation goes from one end of the tapping range to the other, and back again);

b. with the transformer de-energized, and with the auxiliary voltage reduced to 85% of its rated value, one complete cycle of operation;

c. with the transformer energized at rated voltage and frequency at no load, one complete cycle of operation;

d. with one winding short-circuited and, as far as practicable, rated current in the tapped winding, 10 cycles of tap-change operations across the range of two steps on each side from where a coarse or reversing changeover selector operates, or otherwise from the middle tapping (the tapchanger will pass 20 times through the changeover position).

Test result: PASS.

4.14 Lightning impulse test (LI, LIC, LIN) (routine test, type test)

Test date: Aug. 13, 2021

Atmospheric conditions of test:

Relative humidity: 60%; Ambient temperature: 31.7°C; Oil temperature: 32.5°C; Air pressure: 102kPa.

Test items and voltage

Withstand terminals	Rated withstand voltage (kV)		Tapping position
	Lightning full wave	Lightning chopped wave	
A, B, C	480	530	A: 1; B: 9b; C: 17
O	325	/	1
a, b, c	75	85	/

Test sequence:

Line terminal:
One negative reduced level full impulse;
One negative rated level full impulse;
One negative reduced level chopped impulse;
Two negative rated level chopped impulse;
Two negative rated level full impulse.
Neutral point :
One negative reduced level full impulse;
Three negative rated level full impulse.

Test records:

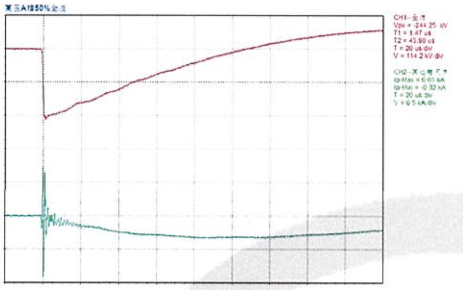
T1: wave front time; T2: time to half peak value; Tc: chopped time; Upk: peak voltage.

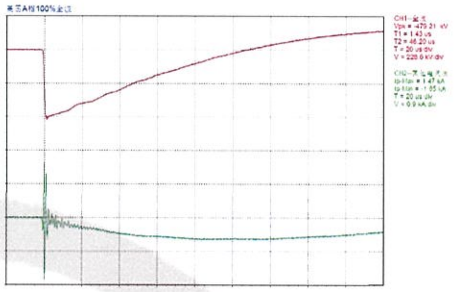
For waveform diagram, see $P_{16\sim22}$.

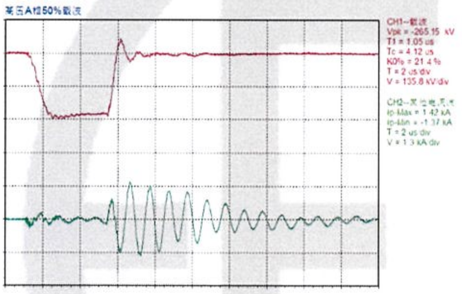
Voltage ranges of oscillograms are as below:

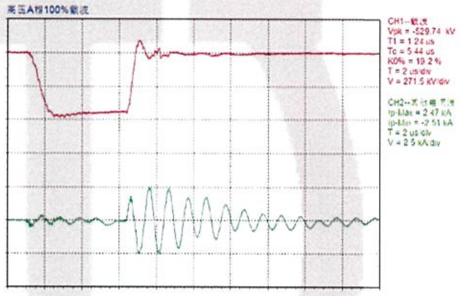
Withstand terminals	Full wave (kV)	Chopped wave (kV)
A, B, C	476.39～481.37	525.29～529.74
O	325.57～327.16	/
a, b, c	74.14～76.43	82.77～85.29

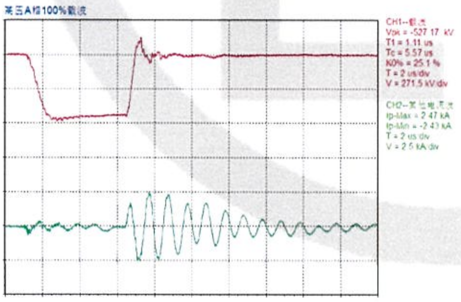
Tested terminal: A Test polarity: negative Channel 1: voltage wave Channel 2: current wave

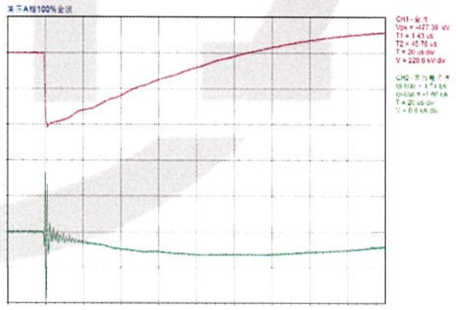


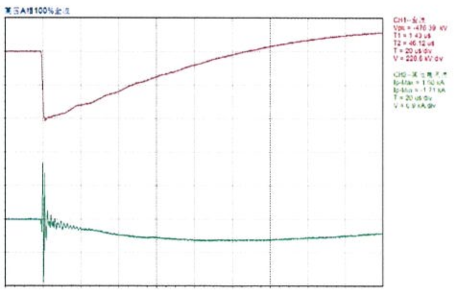




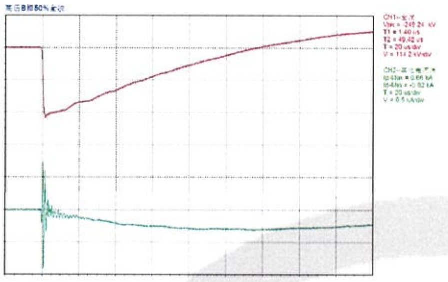


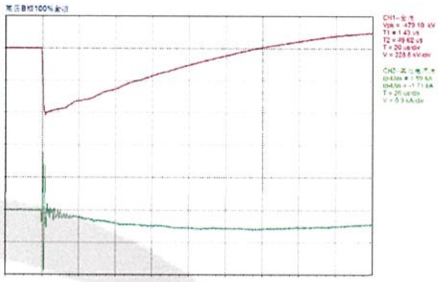


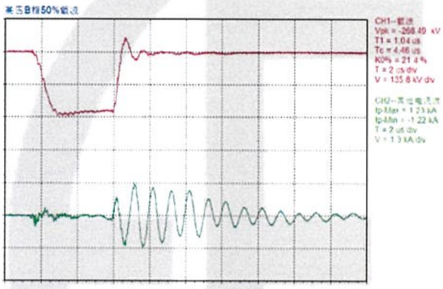


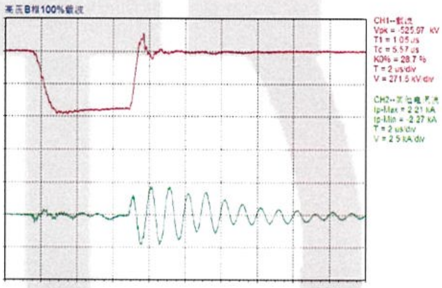


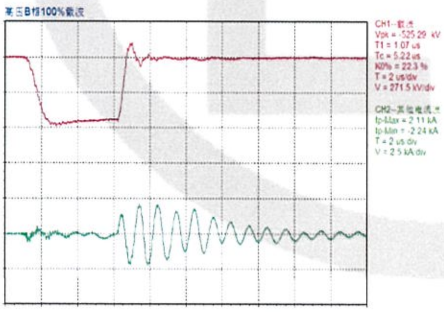
Tested terminal: B   Test polarity: negative   Channel 1: voltage wave   Channel 2: current wave

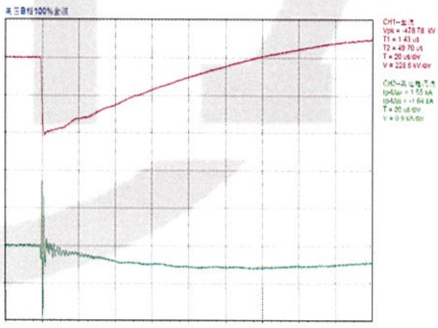


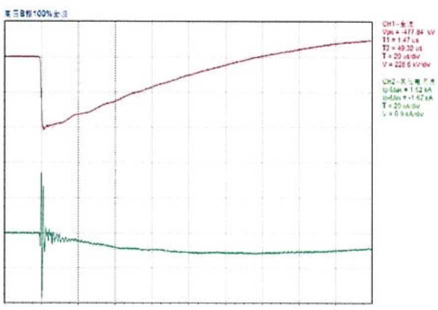




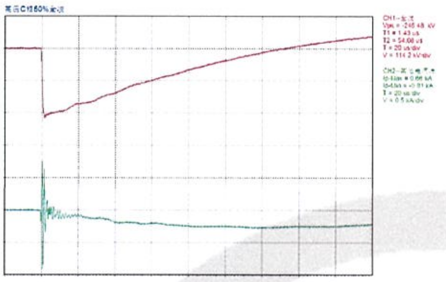


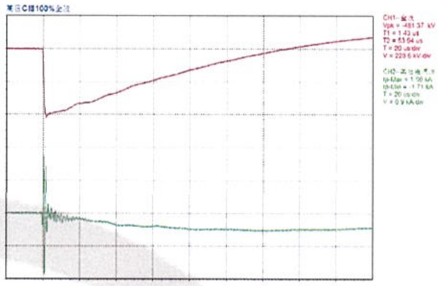


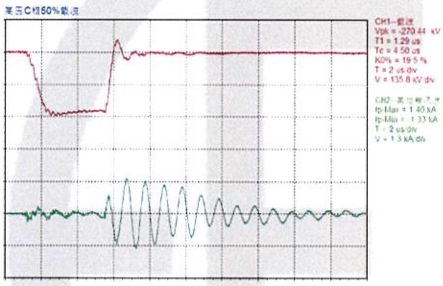


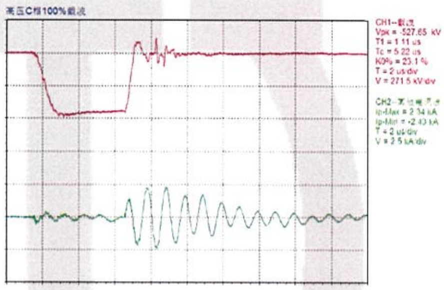


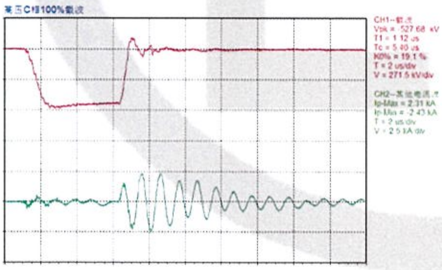
Tested terminal: C Test polarity: negative Channel 1: voltage wave Channel 2: current wave

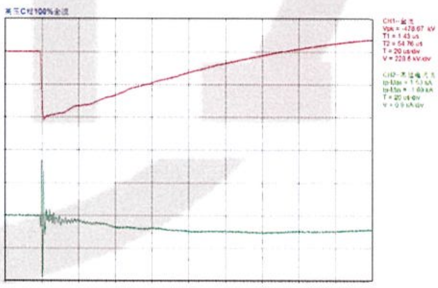


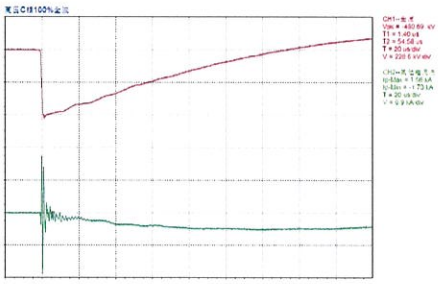




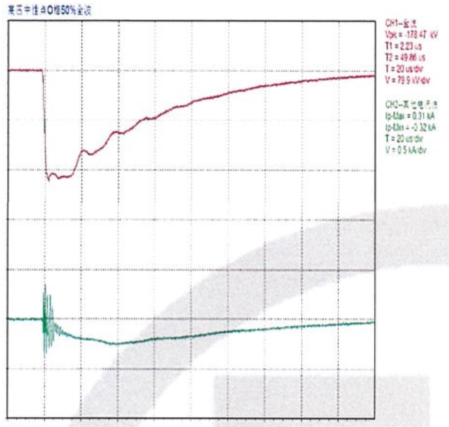


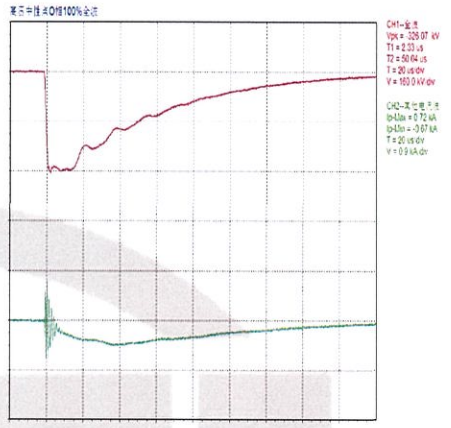


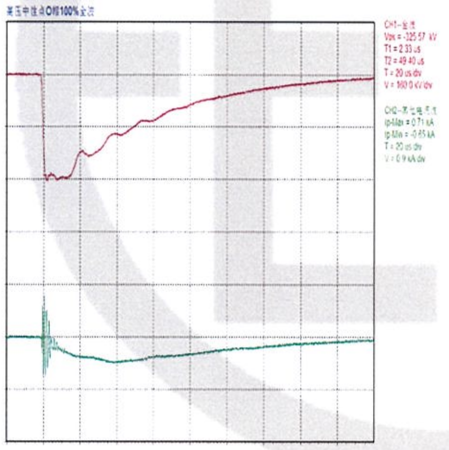


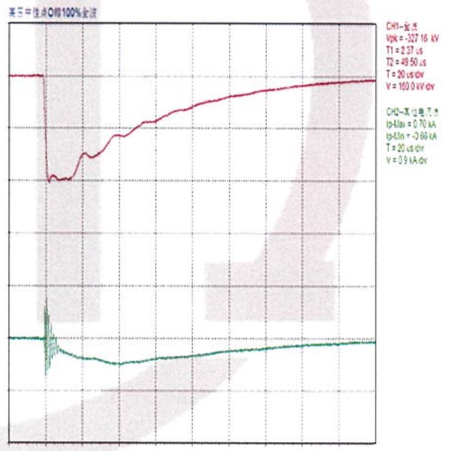


Tested terminal: O Test polarity: negative Channel 1: voltage wave Channel 2: current wave

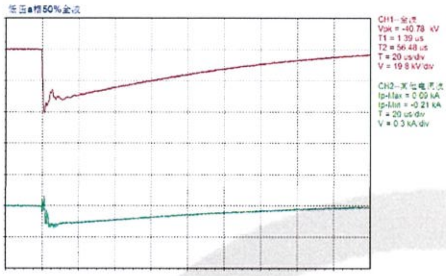


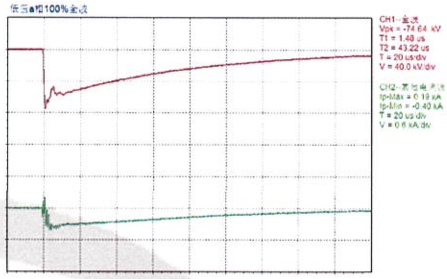


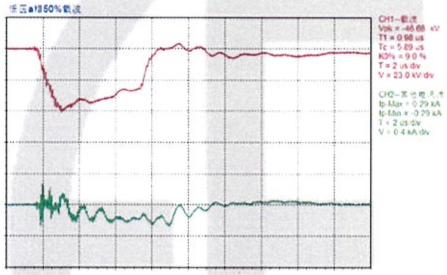


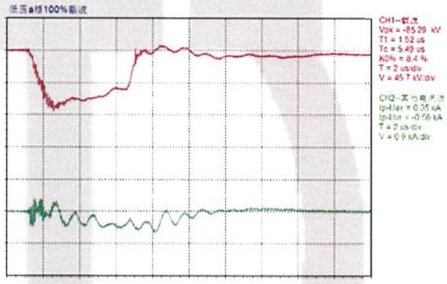


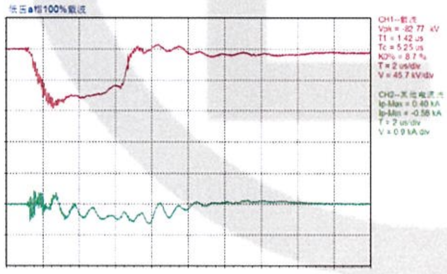
Tested terminal: a Test polarity: negative Channel 1: voltage wave Channel 2: current wave

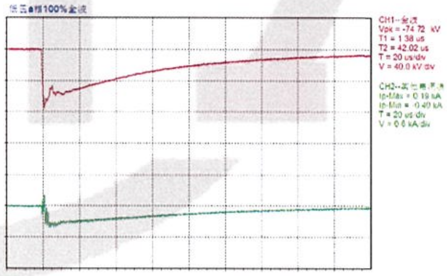


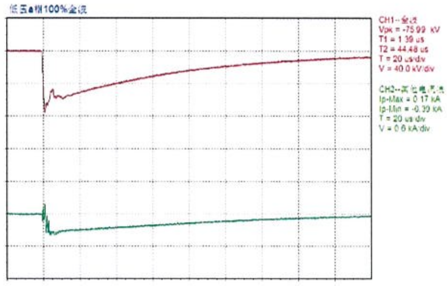




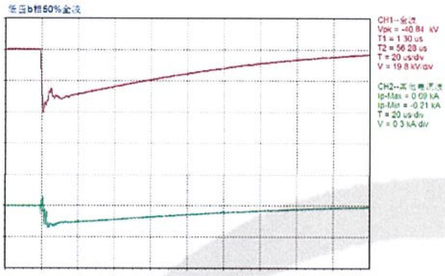


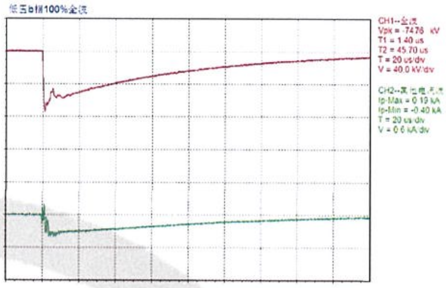


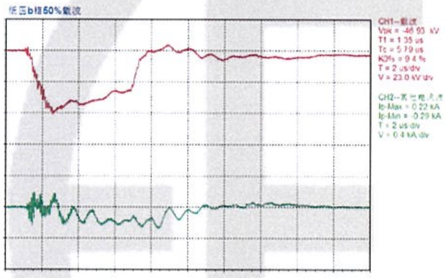


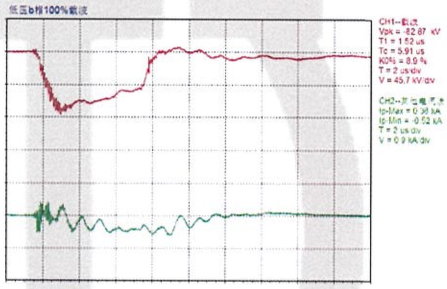


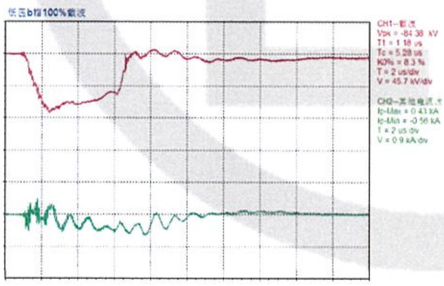
Tested terminal: b Test polarity: negative Channel 1: voltage wave Channel 2: current wave

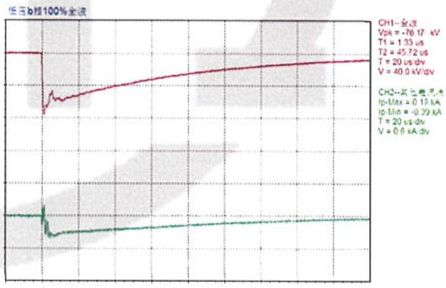


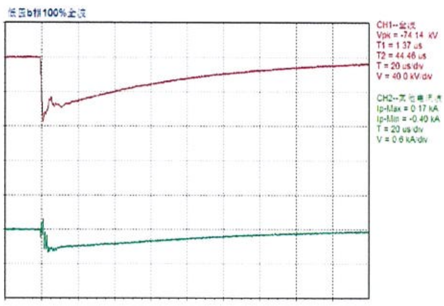




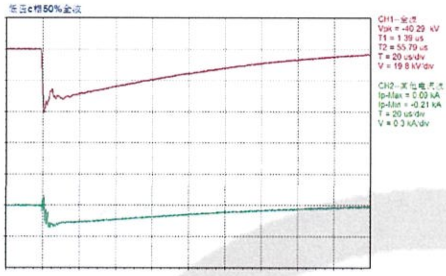


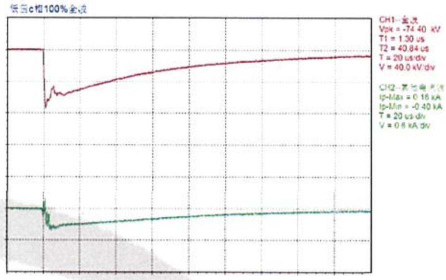


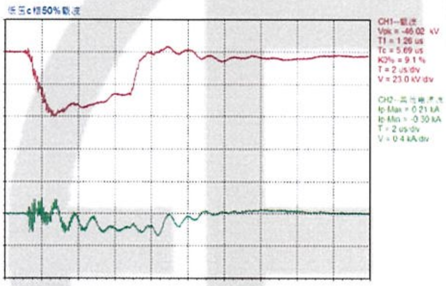


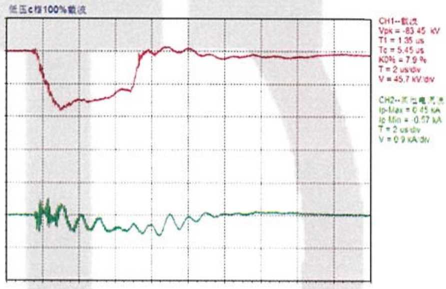


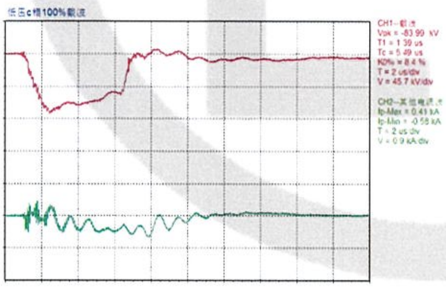
Tested terminal: c   Test polarity: negative   Channel 1: voltage wave   Channel 2: current wave

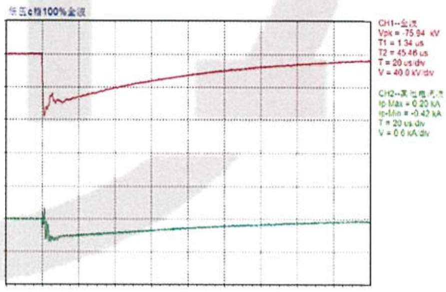


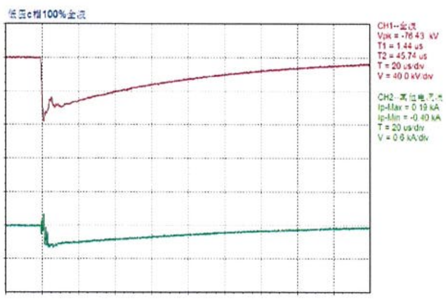












4.15 Applied voltage test (AV) (routine test)

Test date: Aug. 14, 2021

Relative humidity: 67%; Ambient temperature: 32.5°C; Oil temperature: 31.0°C; Air pressure: 102kPa

Applied voltage parts	Test voltage (kV)	Test duration (s)	Result
HV neutral point—LV and earth	140.0	60	PASS
LV—HV and earth	35.0	60	

4.16 Line terminal AC withstand test (LTAC) (routine test)

Test date: Aug. 14, 2021

Relative humidity: 67%; Ambient temperature: 32.5°C; Oil temperature: 32.0°C; Air pressure: 102kPa

Phase to earth test

Applied voltage parts	Tap position	Applied voltage (kV)	Induced voltage (kV)		Frequency (Hz)	Duration (s)	Results
		LV	HV				
a-c	5	21.0	A	200.0	200	30	PASS
a-b		21.0	B	200.0			
b-c		21.0	C	200.0			

4.17 Induced voltage withstand test and induced voltage test with partial discharge measurement (IVW, IVPD) (routine test)

Test date: Aug. 14, 2021

Relative humidity: 67%; Ambient temperature: 32.5°C; Oil temperature: 32.0°C; Air pressure: 102kPa

HV tapping position 9b, frequency 200Hz.

Applied voltage		Duration (min)	Partial discharge magnitude (pC)		
Multiple	Phase-earth voltage (kV)		A	B	C
$0.4Ur/\sqrt{3}$	25.4	/	<30	<30	<30
$1.2Ur/\sqrt{3}$	76.2	1	<40	<40	<40
$1.58Ur/\sqrt{3}$	100.3	5	<50	<70	<50
$2.0Ur/\sqrt{3}$	127.0	0.5	/	/	/
$1.58Ur/\sqrt{3}$	100.3	5	<50	<70	<50
		10	<45	<65	<45
		15	<45	<65	<45
		20	<45	<65	<45
		25	<45	<65	<45
		30	<45	<65	<45
		35	<45	<65	<45
		40	<45	<65	<45
		45	<45	<65	<45
		50	<50	<40	<50
		55	<45	<65	<45
		60	<45	<65	<45
$1.2Ur/\sqrt{3}$	76.2	1	<30	<30	<30
$0.4Ur/\sqrt{3}$	25.4	/	<20	<20	<20

Remark: Ur=110kV.

4.18 Insulating liquid test, measurement of dissolved gasses in dielectric liquid from each separate oil compartment except diverter switch compartment (routine test, type test, commission test)

Test date: Aug. 13, 2021
Relative humidity: 60%; Ambient temperature: 31.7°C

Dielectric dissipation factor (90°C)	Breakdown voltage (kV)	Water content (mg/L)	Flash point (closed cup) (°C)
0.09%	64.3	6.1	173.0

Gas chromatograph analysis (before all tests)

Test date: Aug. 13, 2021
μL/L

H_2	CO	CO_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Total hydrocarbon
0.94	2.85	68.44	0.23	0.00	0.00	0.00	0.23

Gas chromatograph analysis (after dielectric test)

Test date: Aug. 14, 2021
μL/L

H_2	CO	CO_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Total hydrocarbon
0.86	3.87	65.85	0.21	0.00	0.00	0.00	0.21

Gas chromatograph analysis (after long-duration no-load test)

Test date: Aug. 15, 2021
μL/L

H_2	CO	CO_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Total hydrocarbon
1.38	5.42	70.55	0.28	0.00	0.00	0.00	0.28

Gas chromatograph analysis (after temperature-rise test)

Test date: Aug. 15, 2021
μL/L

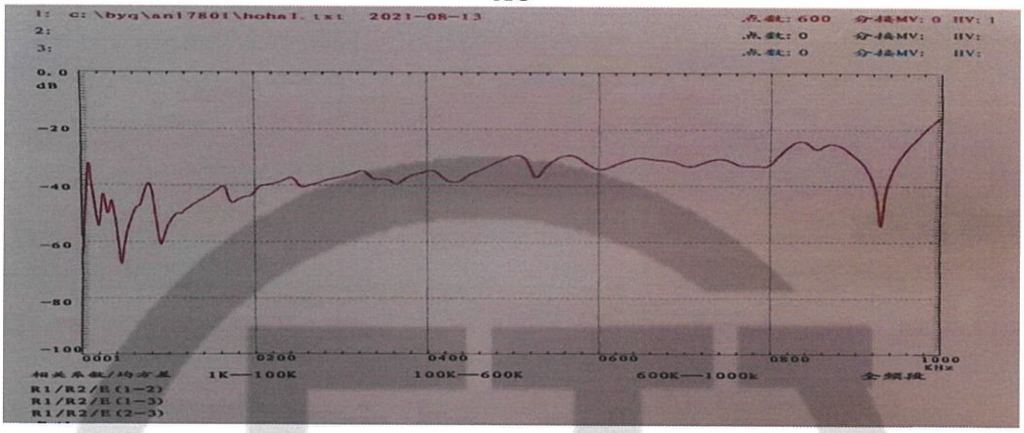
H_2	CO	CO_2	CH_4	C_2H_6	C_2H_4	C_2H_2	Total hydrocarbon
1.46	8.55	98.92	0.32	0.00	0.00	0.00	0.32

4.19 Measurement of frequency response (special test)

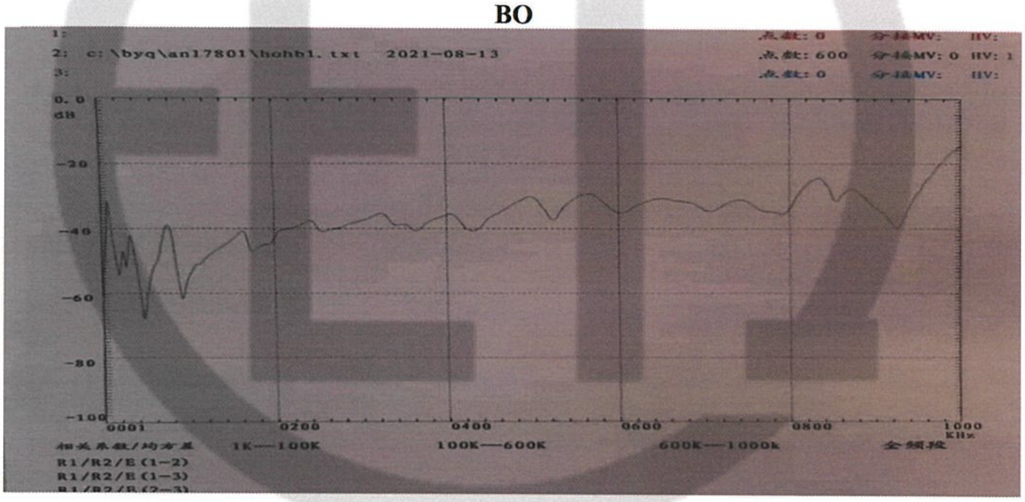
Test date: Aug. 13, 2021

HV winding frequency response curves

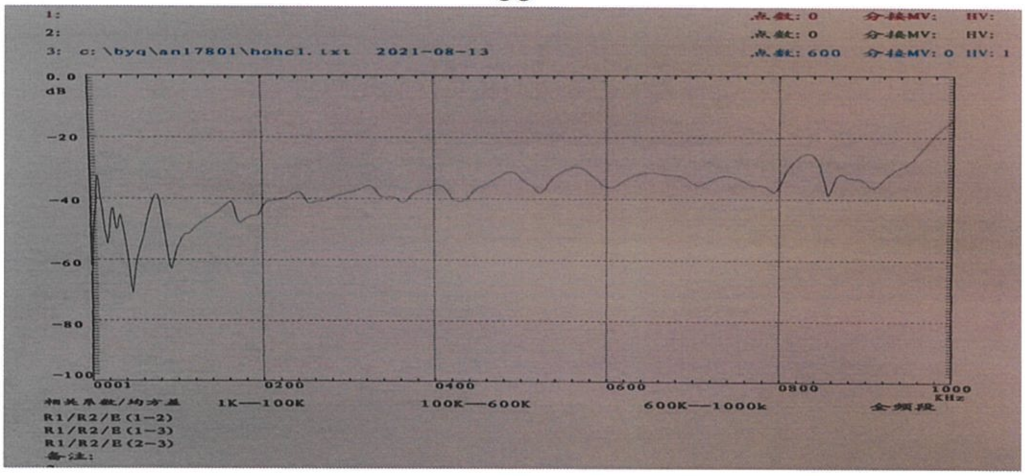
AO



BO

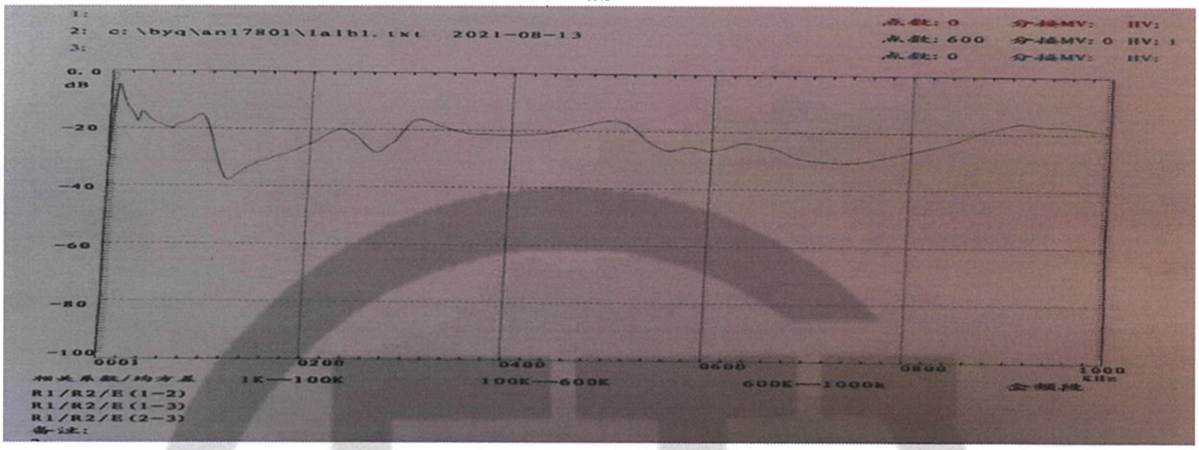


CO



LV winding frequency response curves

ab



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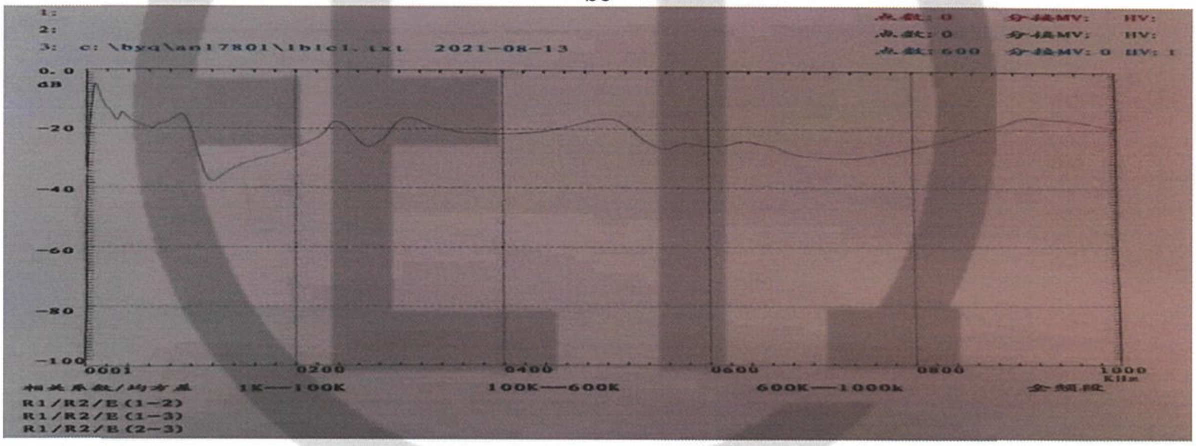

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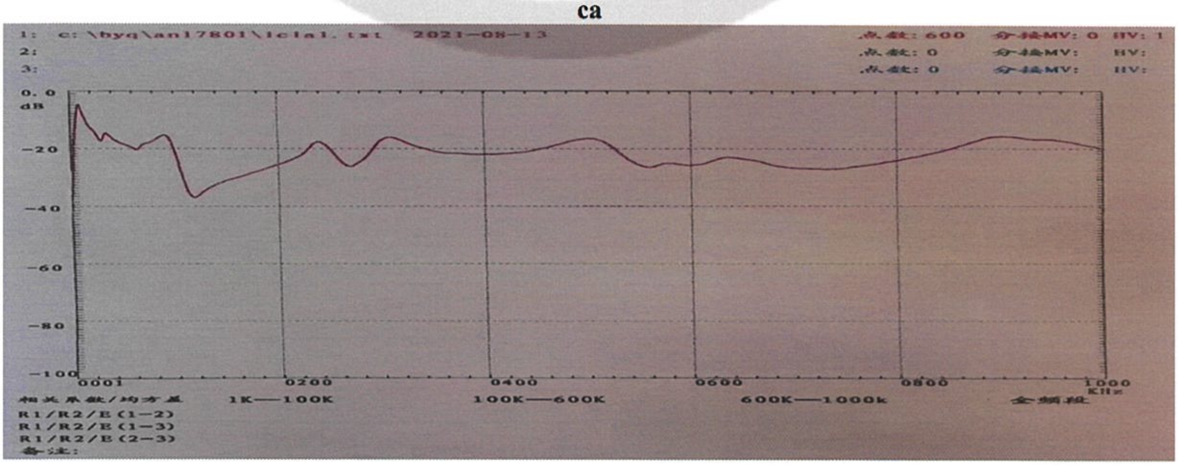
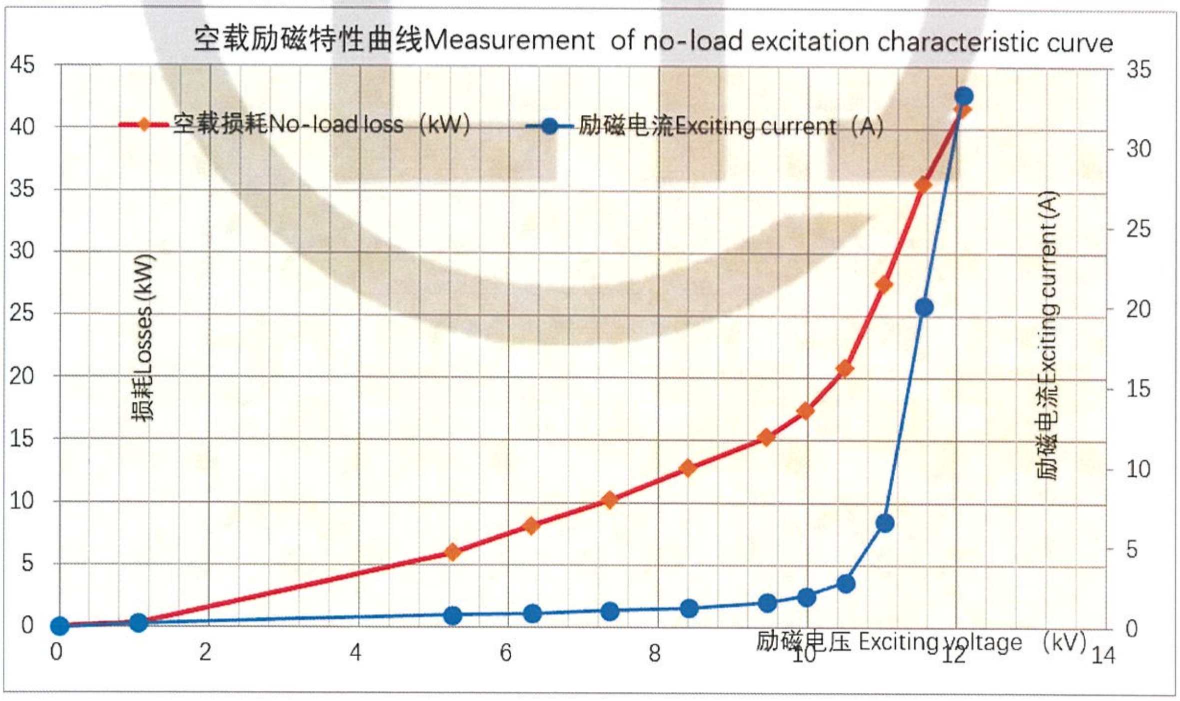


4.20 Measurement of no-load excitation characteristics (commission test)

Test date: Aug. 13, 2021

Voltage	Measured voltage (kV)	Measured current (A)	Measured loss (kW)
$0.10U_r$	1.055	0.216	0.318
$0.50U_r$	5.244	0.709	5.852
$0.60U_r$	6.303	0.822	7.928
$0.70U_r$	7.343	0.957	10.014
$0.80U_r$	8.404	1.134	12.558
$0.90U_r$	9.452	1.143	14.895
$0.95U_r$	9.974	1.725	18.393
$1.00U_r$	10.505	2.353	20.435
$1.05U_r$	11.026	4.427	26.481
$1.10U_r$	11.546	13.979	34.455
$1.15U_r$	12.071	37.508	39.791

Remarks: U_r=10.5kV.



4.21 Long-duration no-load test (commission test)

Test date: from Aug. 14, 2021 to Aug. 15, 2021

Test with 1.1 times rated voltage applied on LV side for 12h. No C_2H_2 is found in oil and the content of total hydrocarbon has no obvious variation before and after the test. For gas chromatograph analysis data, see 4.18.

Duration (h)	Voltage (kV)	Current (A)	No-load loss (kW)
1	11.536	13.85	34.523
2	11.553	13.99	34.495
3	11.548	14.02	34.492
4	11.538	14.03	34.482
5	11.578	14.06	34.492
6	11.557	13.97	34.423
7	11.552	13.98	34.452
8	11.553	13.99	34.484
9	11.552	13.96	34.442
10	11.548	13.89	34.460
11	11.551	13.97	34.452
12	11.556	13.95	34.453

4.22 Measurement of zero-sequence impedances on three-phase transformers (special test)

Test date: Aug. 13, 2021

Connection symbol	Power supply terminal	Open-circuit terminal	Short-circuit terminal	Tapping position	Applied current (A)	Measured voltage (V)	Impedance (Ω)
YNd11	ABC-O	a, b, c	/	9b	102.05	593.93	17.46

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4.23 Measurement of the harmonics of the no-load current (commission test)

Test date: Aug. 13, 2021

No	CH-A THD(%) =18.10		CH-B THD(%) =13.41		CH-C THD(%) =11.40	
	In(A)	In/I1(%)	In(A)	In/I1(%)	In(A)	In/I1(%)
01	0.363	100.000	0.463	100.000	0.267	100.000
02	0.053	14.491	0.031	6.761	0.081	30.195
03	0.062	17.163	0.060	13.003	0.007	2.600
04	0.009	2.371	0.005	1.066	0.010	3.823
05	0.009	2.552	0.012	2.588	0.016	6.023
06	0.008	2.153	0.003	0.569	0.011	4.033
07	0.006	1.795	0.001	0.192	0.008	2.980
08	0.006	1.526	0.002	0.500	0.007	2.744
09	0.004	0.986	0.002	0.490	0.006	2.211
10	0.011	3.036	0.004	0.954	0.007	2.527
11	0.004	1.190	0.003	0.686	0.007	2.673
12	0.005	1.278	0.001	0.165	0.005	2.025
13	0.003	0.827	0.002	0.435	0.005	1.760
14	0.004	0.993	0.002	0.347	0.005	1.744
15	0.003	0.769	0.002	0.353	0.004	1.559
16	0.003	0.741	0.001	0.228	0.004	1.502
17	0.003	0.797	0.001	0.260	0.004	1.498
18	0.002	0.671	0.001	0.218	0.004	1.346
19	0.003	0.717	0.001	0.221	0.003	1.252
20	0.002	0.584	0.001	0.218	0.003	1.076

4.24 Temperature-rise test (including calculation of the winding hot-spot temperature-rise) (type test)

Test date: Aug. 15, 2021

The method of temperature rise is the equivalent test in short-circuit connection, the HV (tapping position 17) is supplying power, the LV is short-circuited, test duration is 12h and of which the stabilization time is 3h.

Measurement of top oil temperature-rise:

During the test, it is required to apply 211.360kW of total loss and 211.680kW of total loss is actually applied.

Measurement of winding temperature-rise:

During the test, 291.60A current is required and 291.46A current is actually applied.

Measured data

Top oil temperature-rise and average oil temperature-rise		Measurement of average temperature-rise windings to oil									Ambient temperature (°C)	
Top oil temperature (°C)	Average oil temperature (°C)	Total loss injection/ specified total loss (%)	Current injection/ rated current (%)	Cold resistance (Ω)		Average oil temperature (°C)		Average winding temperature (°C)			Total loss	Measurement of cold resistance
80.0	66.1	100.15	99.95	HV	0.3534	At the instant of shutdown	64.2	HV	84.8		32.6	30.0
				LV	4.794×10^{-3}	At the end of cooling	59.5	LV	85.9			

Calculations of temperature-rise

Top oil temperature-rise (K)		47.3
Winding temperature-rise (K)	HV	54.2
	LV	55.3
Winding hot-spot temperature-rise (K)	HV	69.7
	LV	70.5
Temperature-rise of tank surface and mental structural parts (K)		50.1

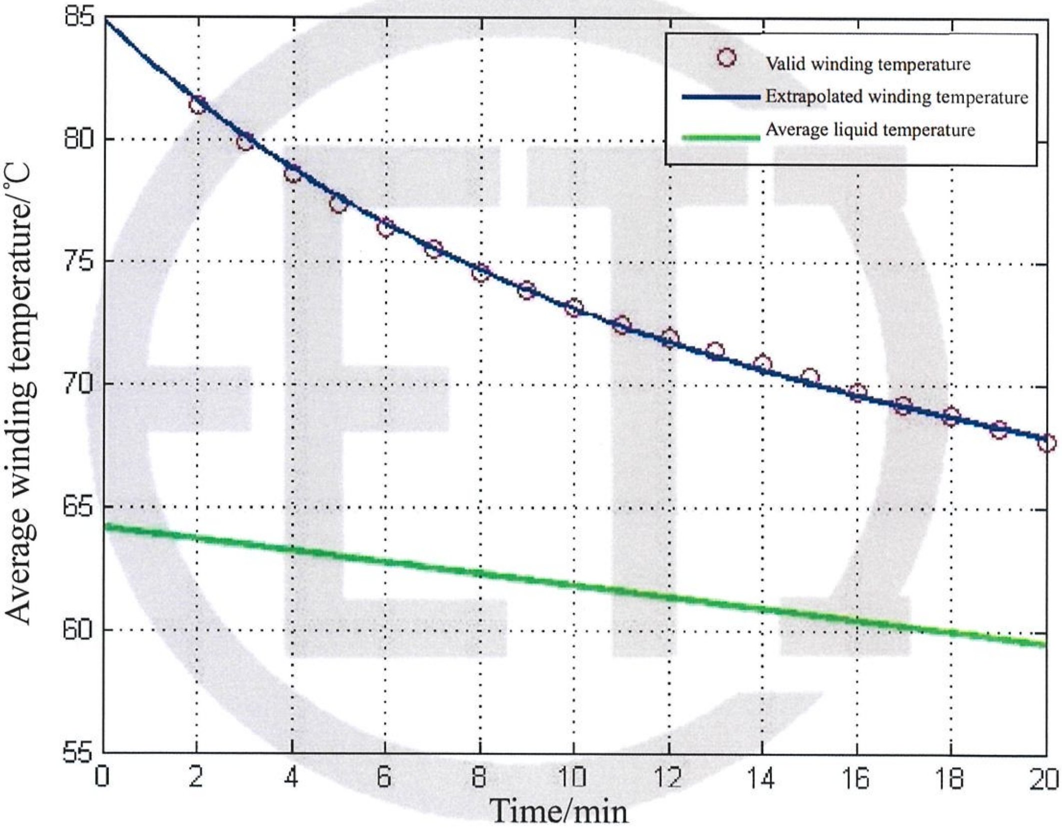
Remarks: the calculated of temperature-rise are the corrected value under specified total loss and rated current. HV winding hot-spot coefficient is 1.08, LV winding hot-spot coefficient is 1.06.

Winding temperature curve

Average winding temperature data

Average HV winding temperature	84.8℃

Average temperature of HV winding at the instant of shut down is 84.83℃

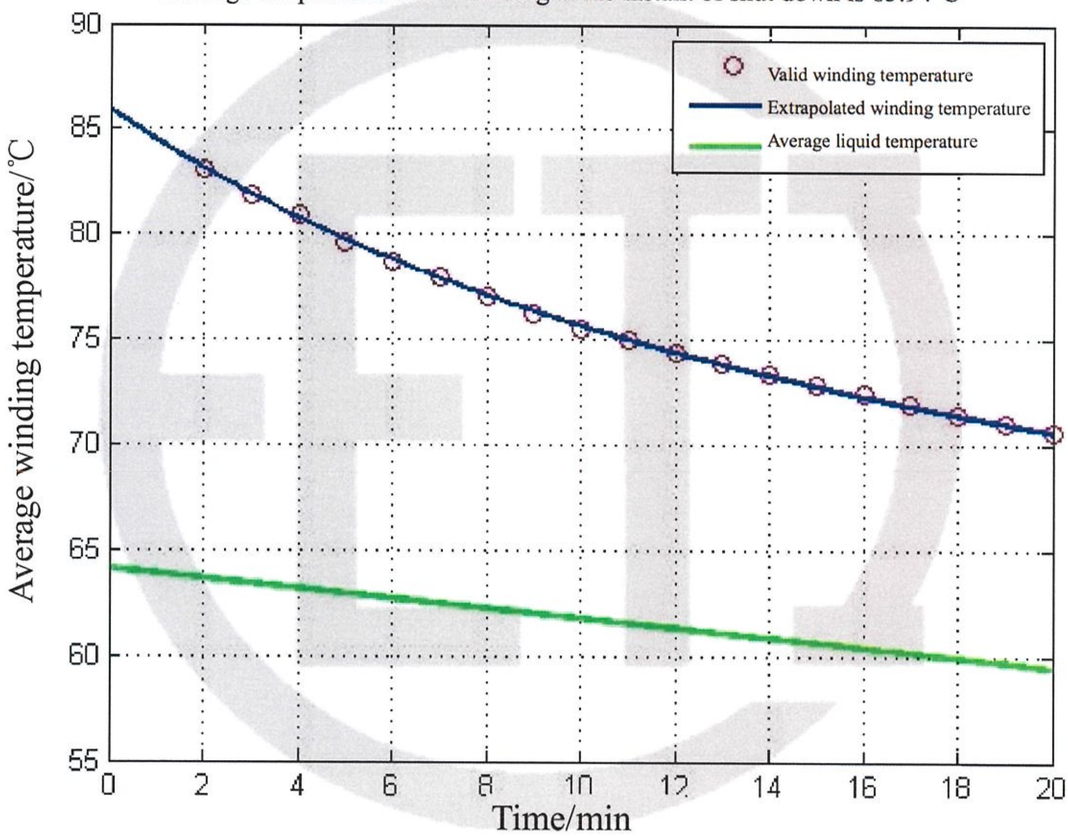


Winding temperature curve

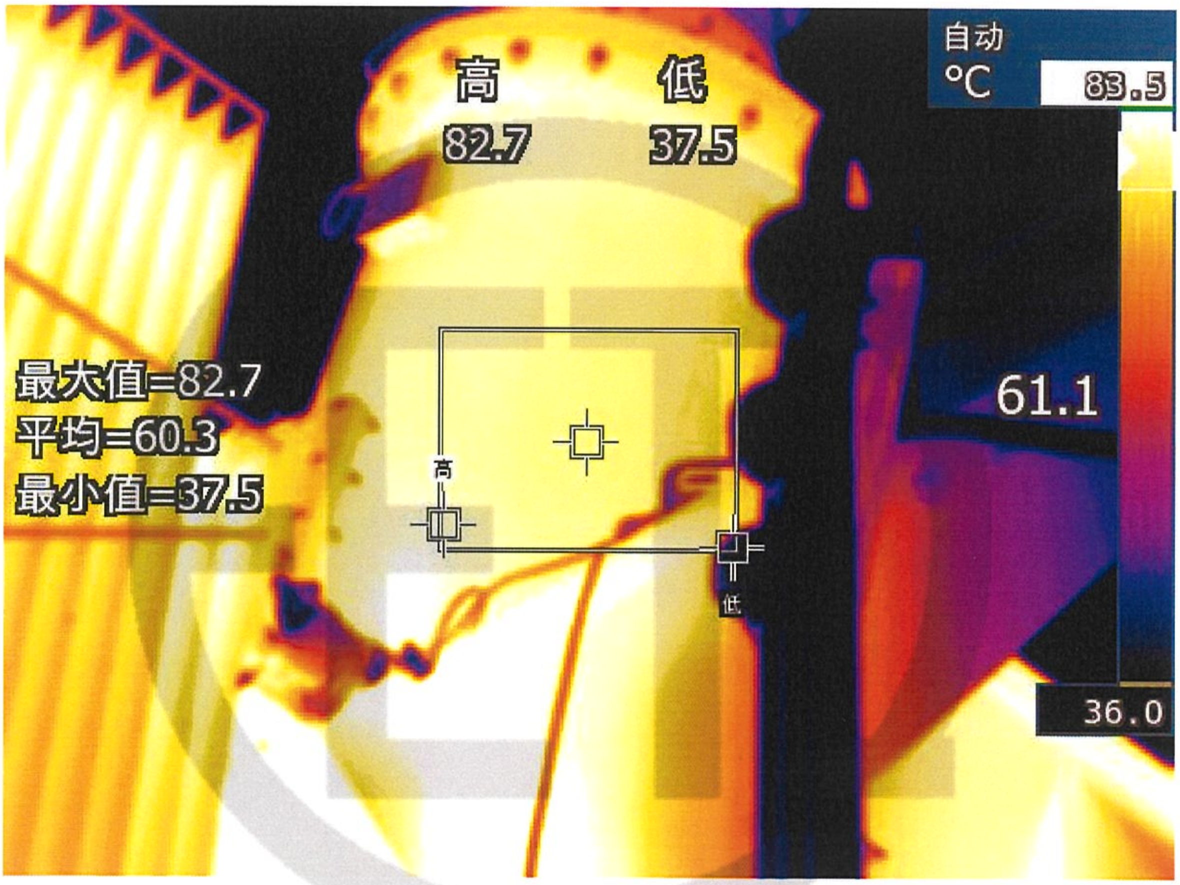
Average winding temperature data

Average LV winding temperature	85.9°C

Average temperature of LV winding at the instant of shut down is 85.94°C



Thermograph of tank surface and structural parts temperature-rise



4.25 Leak testing with pressure for liquid-immersed transformers (routine test)

Test date: from Aug. 10, 2021 to Aug. 12, 2021

Test method	Parts of applied voltage	Applied pressure (kPa)	Duration (h)	Result
Static pressure method	Body	30.0	24	No oil leakage or damage
	On-load tap-changer tank	30.0	24	No oil leakage or damage

Remarks: the product is a general tank, there is no oil leakage or damage before leak testing.

4.26 Determination of sound levels (type test)

Test date: Aug. 14, 2021

4.26.1 Rough estimation of the load current sound power level

Equation:

$$L_{WA,IN} \approx 39 + 18\lg\frac{Sr}{Sp} = 69.6\text{dB(A)}$$

where: Sr—the rated power is 50MVA;
Sp—the reference power is 1MVA.

Because $L_{WA,IN}$ is less 10.4dB(A) than limit value 80dB(A) of assured sound power level, according to standard requirement, it doesn't need to measure load current sound power level.

4.26.2 Measurement of sound pressure level and calculation of sound power level

The transformer is rated excitation, the prescribed contour shall be spaced 0.3m away from the principal radiating surface, the distance between measured points is 0.940m, the number of measured points is 25, the height 1 of measured point is 1.080m, the height 2 of measured point is 2.160m.

Test environment

The total area of the surface of the test room Sv (m^2)	The average acoustic absorption coefficient α	The amount of acoustic absorption A (m^2)	Distance from the principal radiating surface (m)	The area of the measurement surface S (m^2)	Environmental correction K (dB)
5710.0	0.15	856.5	0.3	95.4	1.6

Measured values (dB)

Status of cooling device	Average of background noise		Average noise value of transformer $\overline{L_{PA0}}$	A weighted sound pressure level	A weighted sound power level
	Before the test	After the test		$\overline{L_{pA}} = 10\lg\left(10^{0.1\overline{L_{pA0}}} - 10^{0.1\overline{L_{bgA}}}\right) - K$	$L_{WA} = \overline{L_{pA}} + 10\lg(S/S_0)$
/	43.9	44.0	58.2	56	76

Remarks: $\overline{L_{PA0}}$: the uncorrected average A-weighted sound pressure level; $\overline{L_{PA0}} = 10\lg\left(\frac{1}{N}\sum_{i=1}^{N}10^{0.1L_{PAi}}\right)$

$\overline{L_{bgA}}$: the lower of the two calculated average A weighted background noise pressure level.

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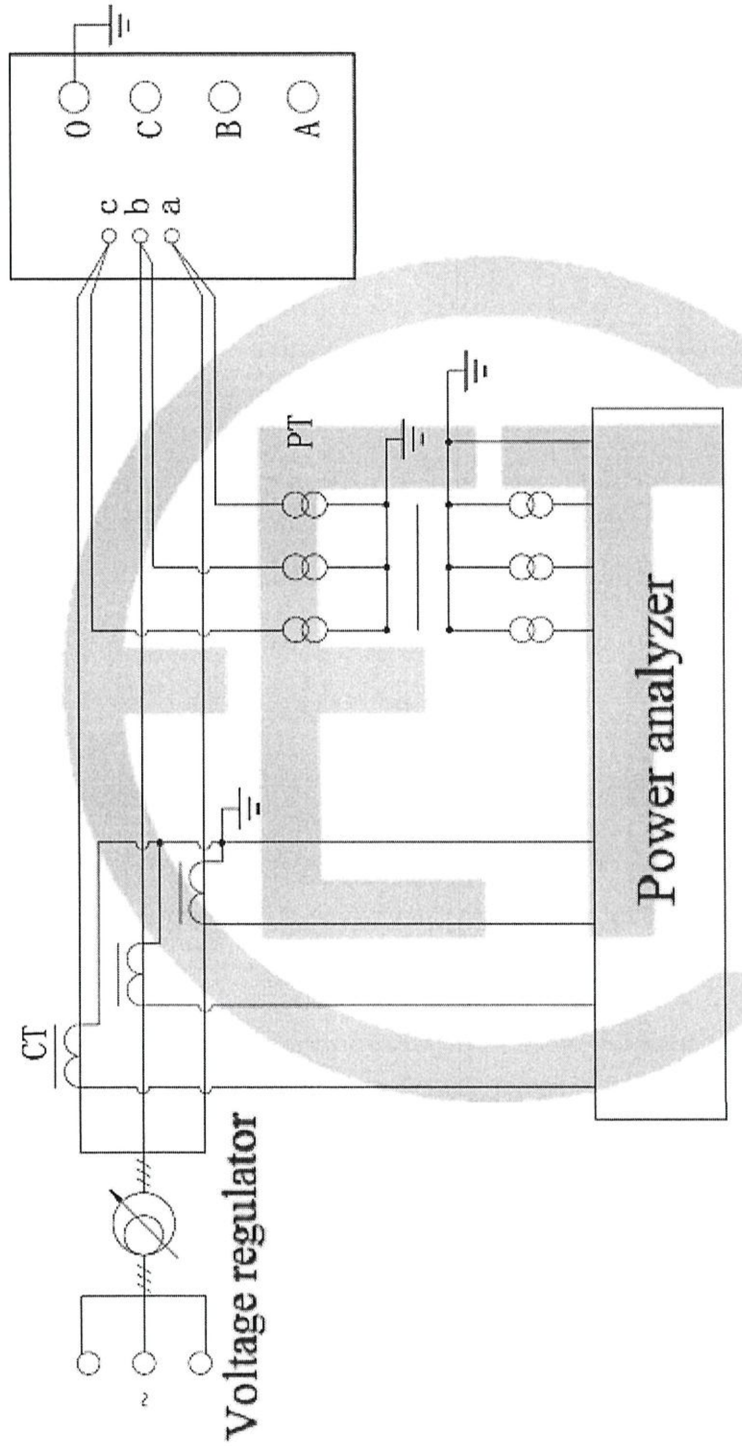
4.27 Mechanical strength test of tank (type test)

Test date: Aug. 17, 2021

Test method	Applied pressure (kPa)	Duration (min)
Vacuum degree	0.133	10
Positive pressure	100	10

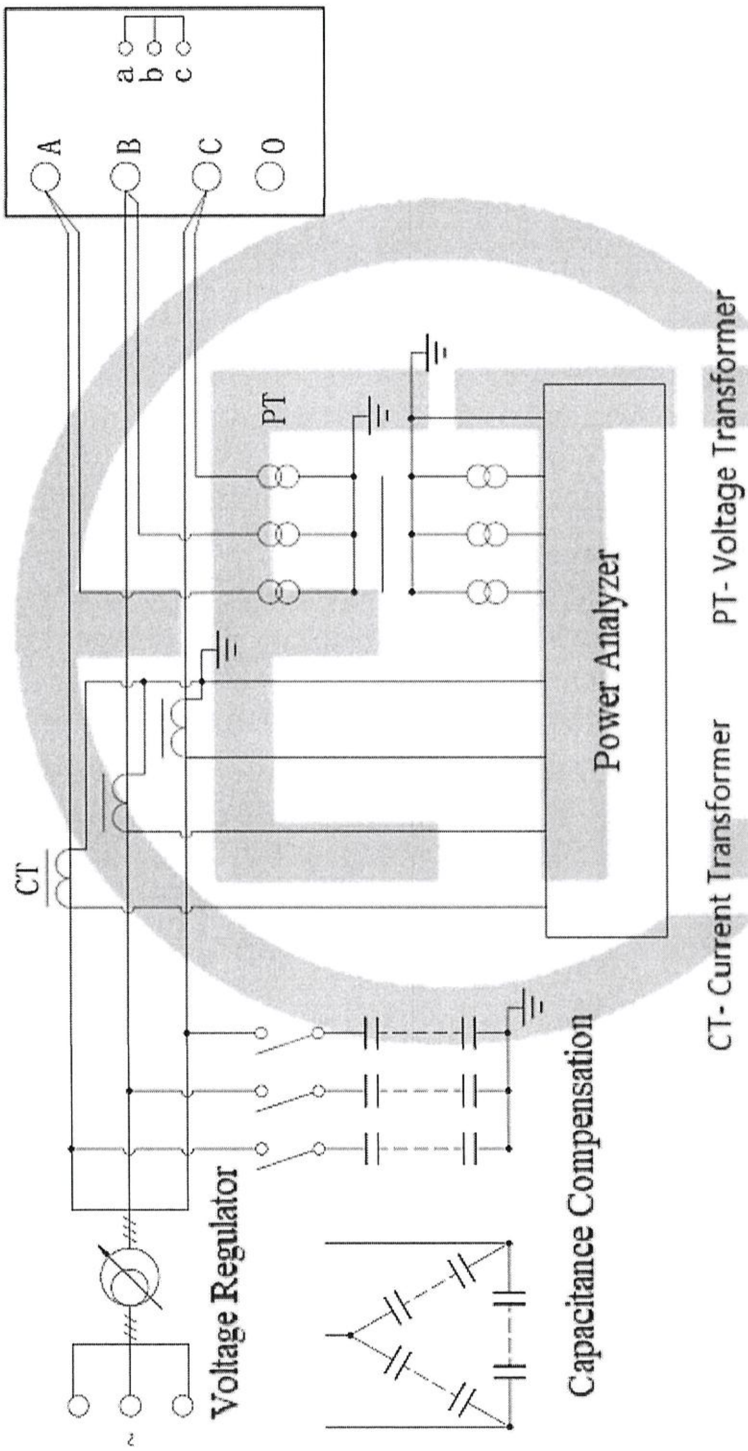
Measured items		Measured points												
		HV side			LV side			Tank cover			Left side		Right side	
Vacuum degree	Initial distance (mm)	205	213	222	242	228	210	280	279	278	211	216	110	99
	Distance after pressure injection (mm)	211	219	226	248	235	221	287	287	285	218	222	116	108
	Distance without pressure (mm)	206	214	222	243	230	212	281	281	278	212	216	111	100
	Elastic deformation (mm)	6	6	4	6	7	11	7	8	7	7	6	6	9
	Permanent deformation (mm)	1	1	0	1	2	2	1	2	1	1	0	1	1
Positive pressure	Initial distance (mm)	206	214	222	243	230	212	281	281	278	212	216	111	100
	Distance after pressure injection (mm)	199	204	213	235	220	203	275	271	272	306	211	104	93
	Distance without pressure (mm)	206	212	221	242	228	211	280	280	278	210	215	111	99
	Elastic deformation (mm)	7	10	9	8	10	9	6	10	6	6	5	6	7
	Permanent deformation (mm)	0	2	1	1	2	1	1	1	0	2	1	0	1
Result		No damage												

Remarks: 1. the product is a general tank.
2. the described left and right sides of test point are viewed from HV side.
3. the left, middle and right test points of HV and LV side are obtained from the 1/2 height in vertical direction, 1/4, 1/2 and 3/4 position respectively in horizontal direction.

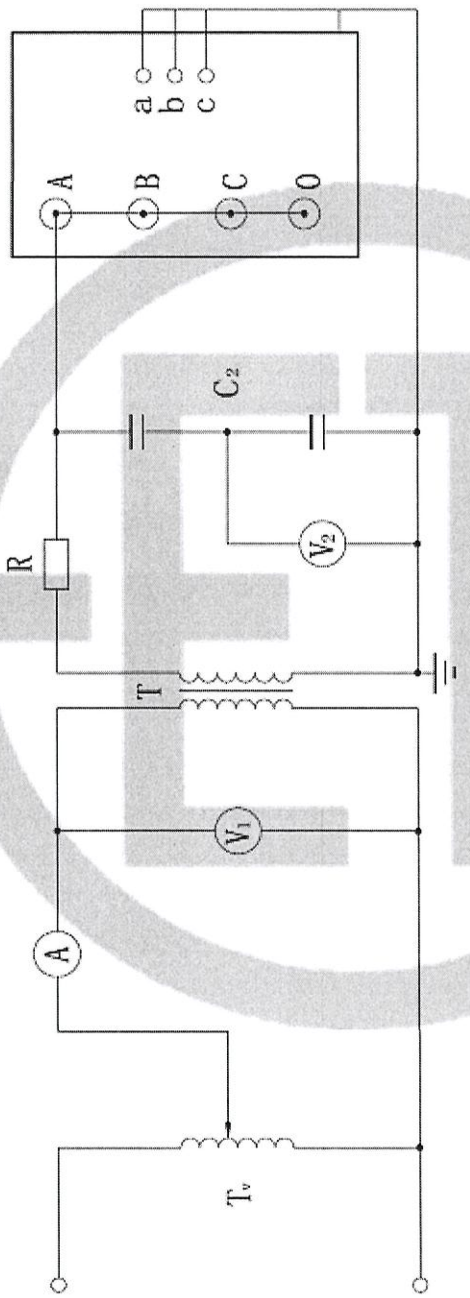


CT- Current transformer PT-Voltage transformer

Schematic diagram of measurement of no-load loss and current

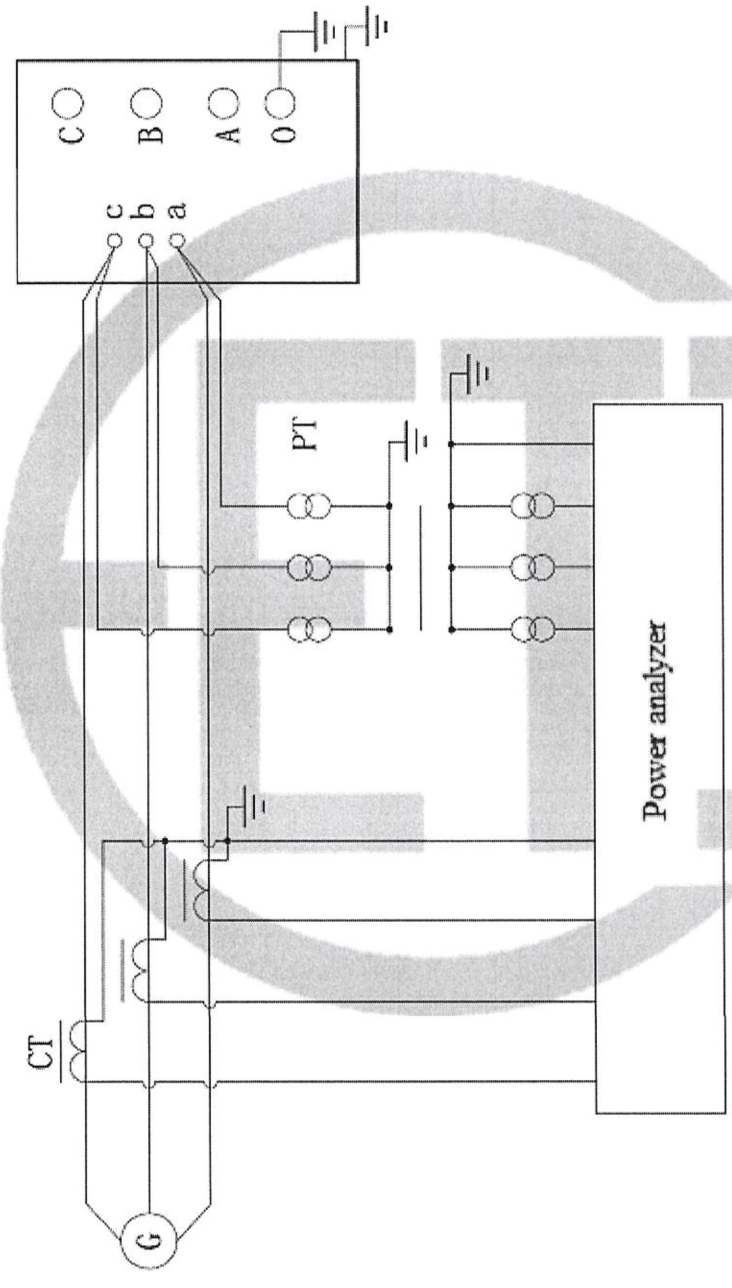


Schematic diagram of measurement of short-circuit impedance and load loss



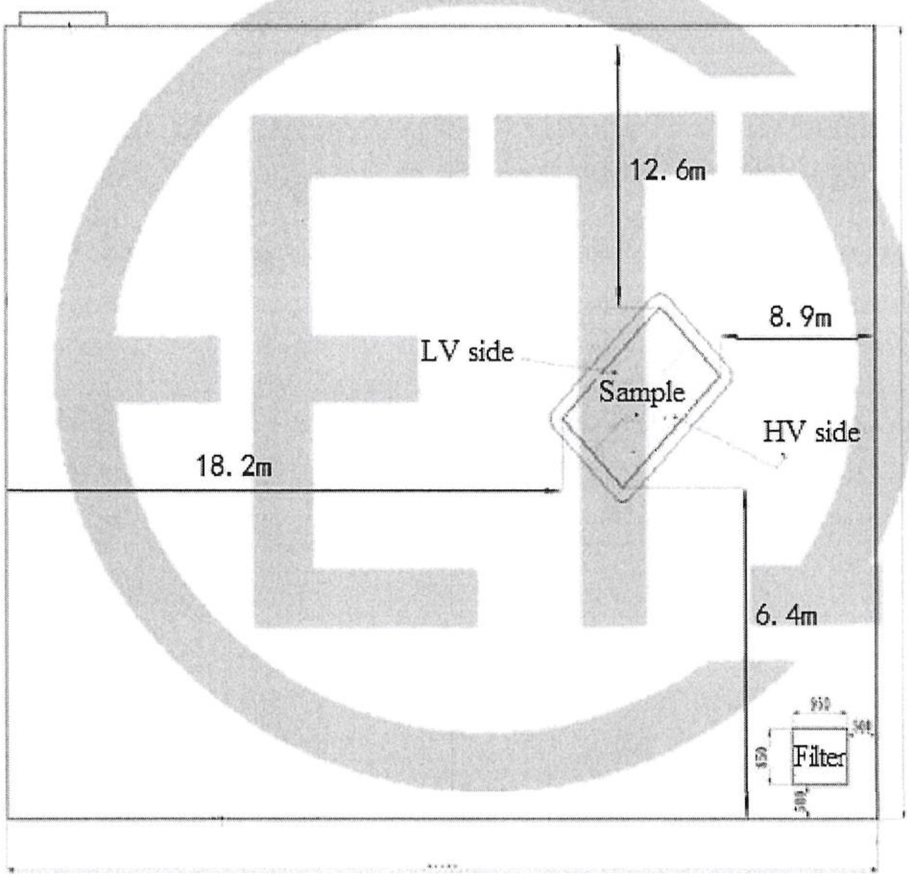
T_v-Power Supply
T-Test Transformer
A- Ammeter
V_1-Voltmeter
R-Resistance
C_2-Capacitor Divider
V_2-Peak Voltmeter

Schematic diagram of applied voltage test



G-Induction power supply CT-Current transformer PT-Voltage transformer

Schematic diagram of induced voltage withstand test



Sound levels arrangement diagram

Instruments used in the tests

No	Test items	Name & type of instrument	Number & validity	Accuracy class
1	Measurement of d.c. insulation resistance windings-to-earth and between windings	Insulation resistance meter JK5053	580576 2022-06-09	/
2	Check of core and frame insulation for liquid-immersed transformers with core or frame insulation			
3	Measurement of dissipation factor (tanδ) of the insulation system capacitances	Full-automatic frequency conversion anti-jamming dielectric dissipation tester JYC	02160521 2022-06-09	/
4	Determination of capacitances windings-to-earth and between windings			
5	Measurement of bushing capacitances and dielectric dissipation factor (tanδ)			
6	Check the ratio and polarity of built-in current transformers	Transformer comprehensive tester JYH-III	1723345 2022-06-10	/
7	Insulation test of auxiliary wiring (AuxW)	AC withstand voltage test system JTKZ	742-152 2022-06-09	/
8	Measurement of voltage ratio and check of phase displacement	Transformer ratio tester JYT-A	04151834 2022-05-09	/
9	Measurement of winding resistance	DC resistance tester JYR40E	01157965 2022-06-09	/
10	Measurement of short-circuit impedance and load loss	Power analyzer WT3000	91N510744 2022-06-09	/
11	Measurement of no-load loss and current			
12	Measurement of no-load loss and current at 90% and 110% of rated voltage			
13	Tests on on-load tap-changers	Transformer on-load tap-changers comprehensive tester BYKC-3310	1723350 2022-06-09	/
14	Lightning impulse test	Impulse voltage test system RZF-1800	1512391 2021-11-09	/
15	Applied voltage test	Power-frequency voltage test system TAWF-1000/300	1512153 2021-11-09	/
16	Line terminal AC withstand test	Power analyzer WT3000	91N510744 2022-06-09	/
17	Induced voltage withstand test and induced voltage test with partial discharge measurement	Power analyzer WT3000	91N510744 2022-06-09	/
		TWPD-2FMulti-channel digital partial discharge comprehensive tester TWPD-2F, eight-channel	2105XY121 2022-05-31	/

Instruments used in the tests

№	Test items	Name & type of instrument	Number & validity		Accuracy class
18	Insulating liquid test, measurement of dissolved gasses in dielectric liquid from each separate oil compartment except diverter switch compartment	Insulating oil dielectric dissipation and resistivity tester HGJD206	1051115	2022-06-09	/
		Dielectric strength tester for insulating oil JKJQ-3	381703415	2021-10-12	/
		Microscale moisture tester CA-2000	E1MB6317	2022-06-09	/
		Gas chromatogram analyzer 2000B	27461380	2022-06-09	/
		Automatic closed flash point tester	JKBS-3	2022-06-09	
19	Measurement of frequency response	Transformer winding deformation tester CS-1B	121028	2022-06-09	/
20	Measurement of no-load excitation characteristics	Power analyzer WT3000	91N510744	2022-06-09	/
21	Long-duration no-load test	Power analyzer WT3000	91N510744	2022-06-09	/
22	Measurement of zero-sequence impedances on three-phase transformers	Power analyzer WT3000	91N510744	2022-06-09	/
23	Measurement of the harmonics of the no-load current				
24	Temperature-rise test	Power analyzer WT3000	91N510744	2022-06-09	/
		Multichannel temperature inspector DT1000	11100033	2022-05-10	/
		Electronic stopwatch SJ9-1	22725	2021-11-10	/
		DC resistance tester JYR40E	01157965	2022-06-09	/
		Infrared thermal camera BritIR	BI30378Y	2022-03-10	
25	Leak testing with pressure for liquid-immersed transformers	Pressure gauge (0~0.1)Mpa	HC69022277554	2021-09-23	/
26	Determination of sound levels	Power analyzer WT3000	91N510744	2022-06-09	/
		Sound level meter HS5633	09016098	2022-04-24	/
27	Mechanical strength test of tank	Pressure gauge (0~0.1)Mpa	HC69022277554	2021-09-23	/
		Stainless steel ruler (0-300) mm/1mm		2022-08-15	/
		Digital vacuum gauge (0~1000) mbar/0.01mbar	2011-11	2022-01-07	/

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The test report is in total 45 pages including 18 figures and 1 photo

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